

JAMES DAO

Houston, TX

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Education

University of Texas at Dallas

Jan 2024 - Aug 2025

Master of Science in Computer Engineering - GPA 3.6

University of Texas at Dallas

Aug 2020 - Dec 2023

Bachelor of Science in Computer Engineering - GPA 3.5

Experiences

Institute of Electrical and Electronics Engineers

Aug 2023 – Aug 2025

Computational Intelligence Society Director

Dallas, TX

- Established a new society branch, increasing student engagement and interest by 25%.
- Led the planning and execution of 5+ industry-level events, attended by 100+ participants, focused on machine learning technologies.
- Developed workshops that increased participants' ML skills, with a 25% improvement in post-workshop assessments.

Cognizant Generative AI Externship

June 2024 – Aug 2024

Student Extern

Dallas, TX

- Gained fundamental knowledge on deep learning, from perceptions to transfer learning, and received hands-on experience with PyTorch and Hugging Face libraries for developing generative models.
- Applied parameter-efficient fine-tuning techniques, resulting in a 30% reduction in computational resource usage for model training.

Multimodal Interaction Lab

Aug 2023 – Dec 2023

Research Intern

Dallas, TX

- Designed and prototyped virtual reality interfaces, enhancing user experience with multisensory feedback. The interfaces increased user engagement by 15%.
- Collaborated with a cross-functional team to develop interaction techniques in Unity using C#, integrating tactile and haptic feedback mechanisms.
- Integrated Unity scenes with real-time sensor data input pipelines, enabling dynamic environmental responses and laying the groundwork for adaptive VR feedback systems.

Projects

Keyboard PCB Design | *KiCad*

- Designed and developed a custom USB-C mechanical keyboard PCB using KiCad, integrating an RP2040 microcontroller with a fully routed switch matrix and power delivery system.
- Implemented USB 2.0 communication and power regulation using a 3.3V LDO, ESD protection, and proper decoupling to ensure stable device operation.
- Optimized PCB layout by transitioning from a 2-layer to 4-layer design, improving signal integrity, grounding, and routing efficiency for a production-ready board.

Facial Recognition IoT System | *Python, Edge ML, IoT, AWS*

- Designed and deployed an edge-based facial recognition system using camera-equipped IoT devices, enabling real-time identity verification with low-latency inference.
- Optimized model performance for resource-constrained environments, reducing inference latency by 30% while maintaining high recognition accuracy.
- Integrated device-to-cloud communication for logging and alerting using AWS services, enabling remote monitoring and scalable system management.

Technical Skills

Languages: Java, Python, C++, C, Verilog

Developer Tools: KiCad, Arduino IDE, Git, VS Code, Docker

Hardware & Systems: PCB Design, Digital Logic Design, Microcontrollers (Arduino, Raspberry Pi), Embedded Systems, Circuit Design,

Protocols & Interfaces: USB 2.0, UART, I2C, SPI

Databases: MongoDB, Firebase, SQL-based Systems, Power Regulation